Computer Science, Paper 1 topics: Computer systems

Mark scheme

Using this mark scheme. A slash (/) separates alternative answers that each score. Words in brackets are not needed for the mark. Give the mark for any answer that means the same as the one listed. In calculations, give full marks for the correct answer with no working, and a method mark if the working is right but the answer is wrong.

Question	Answer	Marks	Guidance
1(a)	To fetch, decode and execute instructions / to process data and instructions	**1**	Do not accept "it stores data" or "it's the brain of the computer" on its own.
1(b)	Program Counter (PC)	**1**	Only one option chosen.
1(c)	Each core can process (fetch, decode and execute) instructions independently (1) so more instructions can be processed at the same time / tasks can be shared between cores / run in parallel (1)	**2**	Accept a correct point that the program must be written to use more than one core as a second mark only if linked to performance.
1(d)	Any one, for example: washing machine / microwave / dishwasher / car engine management system / central heating controller / smart thermostat / traffic lights / digital camera	**1**	Do not accept a general-purpose computer such as a laptop, desktop or tablet.
2(a)	RAM is volatile (loses its contents when the power is off) but ROM is non-volatile (1) RAM can be read from and written to but ROM is read-only / RAM holds the programs and data in use but ROM holds the start-up (boot) instructions / RAM is usually larger than ROM (1)	**2**	Each mark needs both sides of the difference, or one side clearly implying the other. Max 2.
2(b)	182	**1**	
2(c)	01001011	**1**	The question asks for 8 bits, so the leading 0 must be there: 1001011 on its own gets no mark.
2(d)	63	**1**	
2(e)	Answer: 10100001 (2 marks) If the answer is wrong, 1 mark for correct working with the carries shown, or a correct denary check (108 + 53 = 161)	**2**	A correct answer on its own, with no working, gets both marks.
2(f)	Answer: 5000 bytes (2 marks) If the answer is wrong, 1 mark for 100 x 50 x 8 = 40 000 bits, or for dividing their number of bits by 8	**2**	A correct answer on its own, with no working, gets both marks. Accept 5 KB (5000 ÷ 1000). Do not accept 40 000 bytes: that is the number of bits.

2(g)	Lossy compression permanently removes some data (1) but lossless compression removes no data, so the original file can be rebuilt exactly (1)	**2**	Accept "lossy makes a smaller file" as a second mark only if compared with lossless.
3(a)	A network that covers a small geographical area / one site or building (usually with its own hardware)	**1**	Do not accept "a local network" on its own: it repeats the question.
3(b)	If one device or cable fails, the rest of the network still works (1) because each device has its own connection to the central switch (1)	**2**	Also accept: it's easy to add new devices (1) without disrupting the network (1); or fast with few data collisions (1) because data goes through the switch only to the device it's meant for (1).
3(c)	SMTP	**1**	
3(d)	It looks up / translates the domain name (URL) into its IP address (1) and sends the IP address back to the browser so it can request the web page from the web server (1)	**2**	
4(a)	Sending fake emails or messages that pretend to be from a trusted person or organisation (1) to trick the user into giving personal details, such as passwords or bank details, or into clicking a harmful link (1)	**2**	
4(b)	It checks / monitors the data (packets) coming into and going out of the network (1) and blocks any that don't meet its rules / blocks unauthorised access (1)	**2**	Do not accept "it stops all viruses" or "it removes viruses": that is anti-malware.
4(c)	Identify (1), describe (1), for example: memory management (1): allocates RAM to programs / moves data between RAM and virtual memory (1) multitasking (1): lets more than one program run at a time by sharing CPU time (1) peripheral management (1): uses drivers to communicate with devices such as printers (1) user management (1): controls accounts, passwords and access rights (1) file management (1): lets users create, name, move and delete files and folders (1)	**2**	The description must match the function identified.
5	Any two, 1 mark each: used fairly, lawfully and transparently / used only for the purpose it was collected for / adequate, relevant and limited to what's needed / accurate and kept up to date / not kept longer than necessary / kept secure (protected from loss or unauthorised access)	**2**	Also accept that people have the right to see the data held about them. Do not accept "you need permission to use data" on its own.
	Total	**30**	

Total: 30 marks. As a rough guide for this practice paper only (not exam board grade boundaries): 22 or more is a strong result, 15 to 21 is solid with gaps to fix, under 15 means going back over these topics in the notebook first.